



The
University of
Faisalabad

Database Administration

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DATABASE ADMINISTRATION

Lab Assignment Report

Library Management System

System	Library Management System
Tool Used	MySQL Workbench
Subject	Database Administration
Submitted By	Agha Abdullah Khan
Roll No.	2023-BS-IT-28
Submitted To	Sir M.Waleed

1. Project Introduction

1.1 Overview

This lab assignment demonstrates the design and administration of a relational database system for a Library Management System using MySQL Workbench. The project covers all aspects of database administration including schema design, data insertion, login and user management, role-based access control, permission management, and backup and recovery procedures.

1.2 Objectives

- Design and create a normalized relational database with multiple related tables.
- Insert meaningful sample records to simulate real-world library operations.
- Implement login management including creating, modifying, disabling, and enabling logins.
- Connect database users with appropriate logins and permissions.
- Define custom roles and assign users to enforce the principle of least privilege.
- Demonstrate permission granting and revoking with access verification.
- Perform a full database backup and restore from backup after simulated data loss.

1.3 System Description

The Library Management System is designed to manage library operations efficiently. It tracks members who borrow books, maintains a catalog of books and their authors, monitors book loans, and manages library staff. The system consists of five interrelated tables: Members, Authors, Books, Staff, and Loans.

1.4 Tools and Technologies

Tool / Technology	Purpose
MySQL Workbench 8.0+	Database design, query execution, and administration
MySQL Server 8.0+	Relational database engine
mysqldump	Database backup utility (run via terminal)
mysql CLI	Database restore via terminal

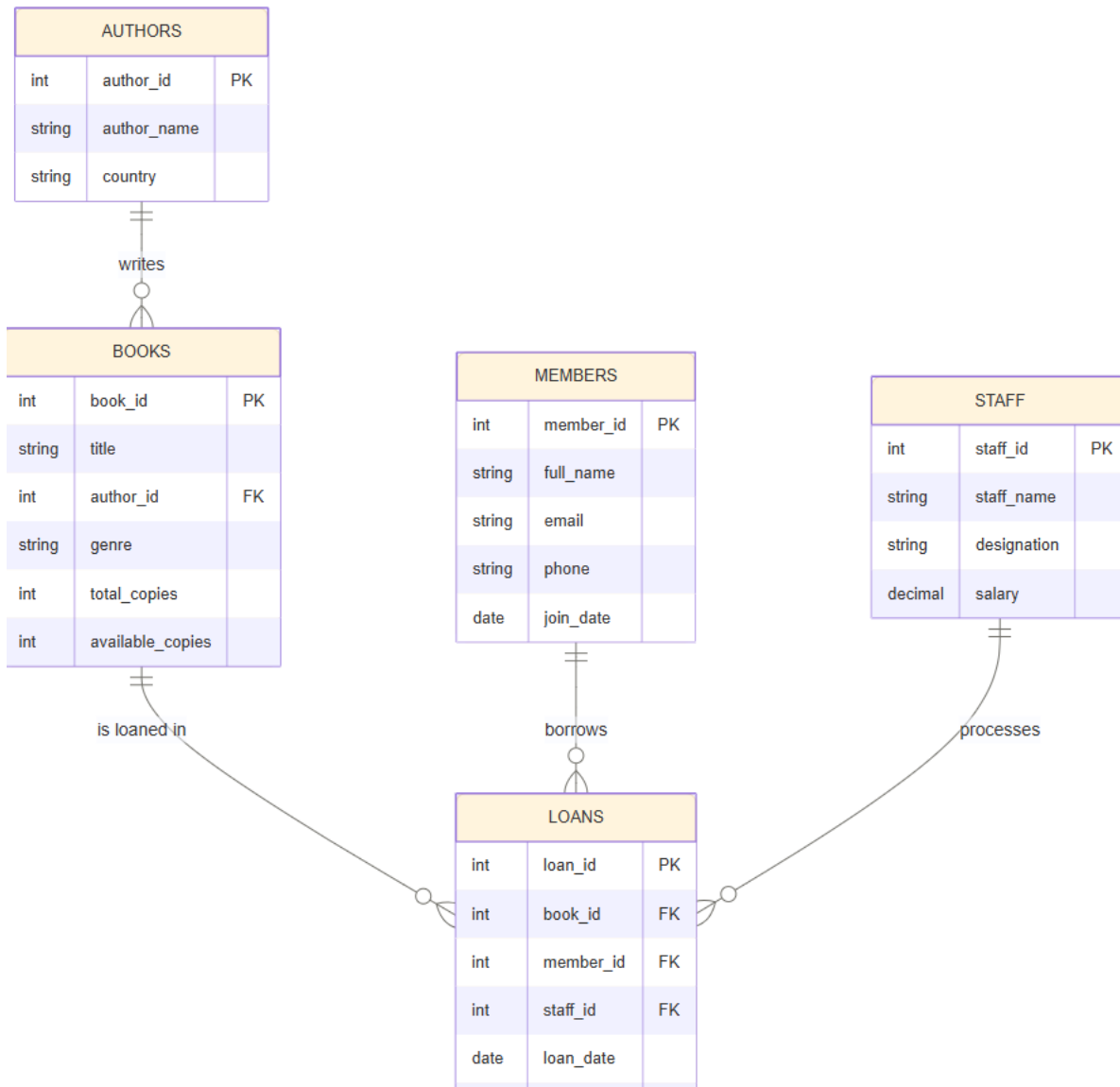
2. Database Diagram (Schema Overview)

The following entity-relationship overview describes the five tables and their relationships:

Table	Primary Key	Foreign Keys	Description
Members	member_id	—	Library members who borrow books
Authors	author_id	—	Authors of the books in the catalog
Books	book_id	author_id → Authors	Books available in the library
Staff	staff_id	—	Library staff who process loans
Loans	loan_id	book_id → Books, member_id → Members, staff_id → Staff	Book loan records

Relationship Summary:

- One Author can write many Books (one-to-many).
- One Member can have many Loans (one-to-many).
- One Book can have many Loans over time (one-to-many).
- One Staff member can process many Loans (one-to-many).

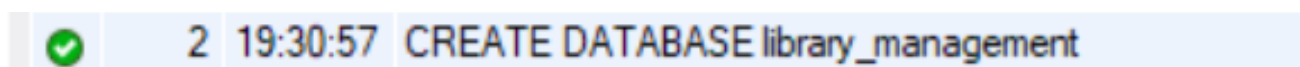


3. Task 1 — Create Database and Tables

3.1 Database Creation

A new database named `library_management` was created using the following SQL commands:

```
DROP DATABASE IF EXISTS library_management;
CREATE DATABASE library_management;
USE library_management;
```



3.2 Table Creation

Five related tables were created to represent the core entities of the library system.

Members Table

```
CREATE TABLE Members (  
    member_id    INT PRIMARY KEY AUTO_INCREMENT,  
    full_name    VARCHAR(100) NOT NULL,  
    email        VARCHAR(100) UNIQUE,  
    phone        VARCHAR(20),  
    join_date    DATE  
);
```

Authors Table

```
CREATE TABLE Authors (  
    author_id    INT PRIMARY KEY AUTO_INCREMENT,  
    author_name  VARCHAR(100) NOT NULL,  
    country      VARCHAR(50)  
);
```

Books Table

```
CREATE TABLE Books (  
    book_id      INT PRIMARY KEY AUTO_INCREMENT,  
    title        VARCHAR(150) NOT NULL,  
    author_id    INT,  
    genre        VARCHAR(50),  
    total_copies INT DEFAULT 1,  
    available_copies INT DEFAULT 1,  
    FOREIGN KEY (author_id) REFERENCES Authors(author_id)  
);
```

Staff Table

```
CREATE TABLE Staff (  
    staff_id     INT PRIMARY KEY AUTO_INCREMENT,  
    staff_name   VARCHAR(100) NOT NULL,  
    designation  VARCHAR(50),  
    salary       DECIMAL(10,2)  
);
```

Loans Table

```
CREATE TABLE Loans (  
    loan_id      INT PRIMARY KEY AUTO_INCREMENT,  
    book_id      INT,  
    member_id    INT,  
    staff_id     INT,  
    loan_date    DATE,  
    return_date  DATE,
```

```

        status          VARCHAR(20) DEFAULT 'Issued',
        FOREIGN KEY (book_id) REFERENCES Books(book_id),
        FOREIGN KEY (member_id) REFERENCES Members(member_id),
        FOREIGN KEY (staff_id) REFERENCES Staff(staff_id)
    );

```

3.3 Sample Data Insertion

	ID	TIME	SQL STATEMENT	MESSAGE
✓	4	19:30:57	CREATE TABLE Members (member_id INT PRIMARY KEY AUTO_INCREM...	0 row(s) affected
✓	5	19:30:57	CREATE TABLE Authors (author_id INT PRIMARY KEY AUTO_INCREMEN...	0 row(s) affected
✓	6	19:30:57	CREATE TABLE Books (book_id INT PRIMARY KEY AUTO_INCREMENT, ...	0 row(s) affected
✓	7	19:30:57	CREATE TABLE Staff (staff_id INT PRIMARY KEY AUTO_INCREMENT, ...	0 row(s) affected
✓	8	19:30:57	CREATE TABLE Loans (loan_id INT PRIMARY KEY AUTO_INCREMENT, ...	0 row(s) affected

Sample records were inserted into all five tables to simulate real library data.

Members inserted (5 records):

```

INSERT INTO Members (full_name, email, phone, join_date) VALUES
('Ali Raza', 'ali.raza@email.com', '03001234567', '2024-01-10'),
('Sara Khan', 'sara.khan@email.com', '03007654321', '2024-02-15'),
('Bilal Ahmed', 'bilal.ahmed@email.com', '03009998888', '2024-03-05'),
('Ayesha Noor', 'ayesha.noor@email.com', '03004445555', '2024-04-20'),
('Hamza Tariq', 'hamza.tariq@email.com', '03002223333', '2024-05-12');

```

Authors inserted (5 records):

```

INSERT INTO Authors (author_name, country) VALUES
('J.K. Rowling', 'UK'), ('George Orwell', 'UK'),
('Paulo Coelho', 'Brazil'), ('Agatha Christie', 'UK'),
('Khaled Hosseini', 'USA');

```

Books inserted (5 records):

```

INSERT INTO Books (title, author_id, genre, total_copies, available_copies) VALUES
('Harry Potter and the Sorcerer's Stone', 1, 'Fantasy', 5, 5),
('1984', 2, 'Dystopian', 4, 4),
('The Alchemist', 3, 'Fiction', 6, 6),
('Murder on the Orient Express', 4, 'Mystery', 3, 3),
('The Kite Runner', 5, 'Drama', 4, 4);

```

Staff inserted (3 records):

```

INSERT INTO Staff (staff_name, designation, salary) VALUES
('Imran Sheikh', 'Librarian', 45000.00),
('Nadia Iqbal', 'Assistant Librarian', 32000.00),
('Usman Farooq', 'Library Clerk', 25000.00);

```

Loans inserted (5 records):

4. Task 2 — Login Management

Three database logins were created with different access levels to represent an Admin, Librarian, and Clerk.

4.1 Create Logins

```
CREATE USER 'login_admin'@'localhost' IDENTIFIED BY 'Admin@123';
CREATE USER 'login_librarian'@'localhost' IDENTIFIED BY 'Lib@123';
CREATE USER 'login_clerk'@'localhost' IDENTIFIED BY 'Clerk@123';
```

	#	Time	Action	Message
✓	20	19:30:57	CREATE USER 'login_admin'@'localhost' IDENTIFIED BY 'Admin@123'	0 row(s) affected
✓	21	19:30:57	CREATE USER 'login_librarian'@'localhost' IDENTIFIED BY 'Lib@123'	0 row(s) affected
✓	22	19:30:57	CREATE USER 'login_clerk'@'localhost' IDENTIFIED BY 'Clerk@123'	0 row(s) affected
✓	23	19:30:57	ALTER USER 'login_clerk'@'localhost' IDENTIFIED BY 'NewClerk@456'	0 row(s) affected
✓	24	19:30:57	ALTER USER 'login_clerk'@'localhost' ACCOUNT LOCK	0 row(s) affected
✓	25	19:30:57	ALTER USER 'login_clerk'@'localhost' ACCOUNT UNLOCK	0 row(s) affected

4.2 Change Password

The password for login_clerk was changed:

```
ALTER USER 'login_clerk'@'localhost' IDENTIFIED BY 'NewClerk@456';
```

4.3 Disable and Enable Login

The login_clerk account was locked (disabled) and then unlocked (enabled):

```
-- Disable:
ALTER USER 'login_clerk'@'localhost' ACCOUNT LOCK;

-- Enable:
ALTER USER 'login_clerk'@'localhost' ACCOUNT UNLOCK;
```

✓	23	19:30:57	ALTER USER 'login_clerk'@'localhost' IDENTIFIED BY 'NewClerk@456'	
✓	24	19:30:57	ALTER USER 'login_clerk'@'localhost' ACCOUNT LOCK	
✓	25	19:30:57	ALTER USER 'login_clerk'@'localhost' ACCOUNT UNLOCK	

5. Task 3 — User Management

In MySQL, the CREATE USER command serves as both the login and user creation step. The following grants connected the three users to the library_management database:

```
GRANT USAGE ON library_management.* TO 'login_admin'@'localhost';  
GRANT USAGE ON library_management.* TO 'login_librarian'@'localhost';  
GRANT USAGE ON library_management.* TO 'login_clerk'@'localhost';
```

The GRANT USAGE statement allows the user to connect to the database without granting any additional privileges. Further permissions are assigned through roles (see Task 4 and Task 5).

✓	26	19:30:57	GRANT USAGE ON library_management.* TO 'login_admin'@'localhost'	0 row(s) affected
✓	27	19:30:57	GRANT USAGE ON library_management.* TO 'login_librarian'@'localhost'	0 row(s) affected
✓	28	19:30:57	GRANT USAGE ON library_management.* TO 'login_clerk'@'localhost'	0 row(s) affected

6. Task 4 — Role Management

6.1 Create Custom Roles

Two custom roles were created to represent different levels of access:

```
CREATE ROLE 'admin_role';
CREATE ROLE 'readonly_role';
```

6.2 Assign Users to Roles

Users were assigned to the appropriate roles based on their job function:

```
GRANT 'admin_role'      TO 'login_admin'@'localhost';
GRANT 'readonly_role' TO 'login_librarian'@'localhost';
GRANT 'readonly_role' TO 'login_clerk'@'localhost';

SET DEFAULT ROLE ALL TO 'login_admin'@'localhost';
SET DEFAULT ROLE ALL TO 'login_librarian'@'localhost';
SET DEFAULT ROLE ALL TO 'login_clerk'@'localhost';
```

Login	Role Assigned	Access Level
login_admin	admin_role	Full database access
login_librarian	readonly_role	Read-only + Loans INSERT
login_clerk	readonly_role	Read-only access only

✓	29	19:30:57	CREATE ROLE 'admin_role'
✓	30	19:30:57	CREATE ROLE 'readonly_role'
✓	31	19:30:57	GRANT 'admin_role' TO 'login_admin'@'localhost'
✓	32	19:30:57	GRANT 'readonly_role' TO 'login_librarian'@'localhost'
✓	33	19:30:57	GRANT 'readonly_role' TO 'login_clerk'@'localhost'

7. Task 5 — Permission Management

7.1 Grant Permissions

Permissions were assigned to roles and users as follows:

Full privileges to admin_role:

```
GRANT ALL PRIVILEGES ON library_management.* TO 'admin_role';
```

Read-only SELECT permission to readonly_role:

```
GRANT SELECT ON library_management.* TO 'readonly_role';
```

Additional INSERT and UPDATE on Loans to login_librarian:

```
GRANT INSERT, UPDATE ON library_management.Loans TO 'login_librarian'@'localhost';
```

```
FLUSH PRIVILEGES;
```

7.2 Revoke a Permission

The UPDATE permission on the Loans table was revoked from login_librarian:

```
REVOKE UPDATE ON library_management.Loans FROM 'login_librarian'@'localhost';
```

7.3 Demonstrate Allowed and Denied Access

User	Query Attempted	Result
login_librarian	SELECT * FROM library_management.Books;	ALLOWED
login_librarian	UPDATE library_management.Loans SET status='Returned'...	DENIED (revoked)
login_clerk	INSERT INTO library_management.Members (...) VALUES (...);	DENIED (readonly_role)
login_admin	All DDL and DML operations	ALLOWED (all privileges)

Check granted privileges with:

```
SHOW GRANTS FOR 'login_admin'@'localhost';  
SHOW GRANTS FOR 'login_librarian'@'localhost';  
SHOW GRANTS FOR 'login_clerk'@'localhost';
```

✓	42	19:30:57	SHOW GRANTS FOR 'login_admin'@'localhost'	2 row(s) returned
✓	43	19:30:57	SHOW GRANTS FOR 'login_librarian'@'localhost'	3 row(s) returned
✓	44	19:30:57	SHOW GRANTS FOR 'login_clerk'@'localhost'	2 row(s) returned

8. Task 6 — Backup and Recovery

8.1 Full Database Backup

A full backup of the library_management database was taken using mysqldump from the terminal (not inside Workbench's query tab):

```
mysqldump -u root -p library_management > library_management_backup.sql
```

This command exports the entire database schema and data into a .sql file that can be used to restore the database.

```
C:\Program Files\MySQL\MySQL Server 8.0\bin>mysqldump -u root -p library_management > library_management_backup.sql
Enter password: *****

C:\Program Files\MySQL\MySQL Server 8.0\bin>dir library_management_backup.sql
Volume in drive C has no label.
Volume Serial Number is 6CC8-1810

Directory of C:\Program Files\MySQL\MySQL Server 8.0\bin

06/17/2026  12:42 AM                6,823 library_management_backup.sql
             1 File(s)                6,823 bytes
             0 Dir(s) 130,149,265,408 bytes free
```

8.2 Simulate Data Loss

Two records were deleted to simulate accidental or intentional data loss:

```
DELETE FROM Loans WHERE loan_id = 5;
DELETE FROM Members WHERE member_id = 5;
```

Verification after deletion:

```
SELECT * FROM Loans;
SELECT * FROM Members;
```

✓ 94 11:40:19 DELETE FROM Loans WHERE loan_id = 5

Result Grid							
		Filter Rows:		Export:		Wrap Cell Content:	
	loan_id	book_id	member_id	staff_id	loan_date	return_date	status
▶	1	1	1	1	2025-01-05	2025-01-19	Returned
	2	2	2	2	2025-01-10	NULL	Issued
	3	3	3	1	2025-02-01	2025-02-15	Returned
	4	4	4	3	2025-02-10	NULL	Issued

99 11:42:50 DELETE FROM Members WHERE member_id = 5

Result Grid |   Filter Rows: | Export:  | Wrap Cell Content: 

	member_id	full_name	email	phone	join_date
▶	1	Agha Abdullah Khan	agha.khan@email.com	03001234567	2024-01-10
	2	Kala jee	kala@email.com	03007654321	2024-02-15
	3	Uzman Ahmad	uzman.ahmad@email.com	03009998888	2024-03-05
	4	Habib Ali	ayesha.noor@email.com	03004445555	2024-04-20

8.3 Restore from Backup

The database was restored from the backup file using the mysql command in the terminal:

```
mysql -u root -p library_management < library_management_backup.sql
```

After restoration, all 5 Members and all 5 Loans records were verified to be present again.

```
C:\Program Files\MySQL\MySQL Server 8.0\bin>cd "C:\Program Files\MySQL\MySQL Server 8.0\bin"
C:\Program Files\MySQL\MySQL Server 8.0\bin>mysql -u root -p library_management < library_management_backup.sql
Enter password: *****
C:\Program Files\MySQL\MySQL Server 8.0\bin>
```

Result Grid |   Filter Rows: | Export:  | Wrap Cell Content: 

	member_id	full_name	email	phone	join_date
▶	1	Agha Abdullah Khan	agha.khan@email.com	03001234567	2024-01-10
	2	Kala jee	kala@email.com	03007654321	2024-02-15
	3	Uzman Ahmad	uzman.ahmad@email.com	03009998888	2024-03-05
	4	Habib Ali	ayesha.noor@email.com	03004445555	2024-04-20
	5	lightswali	lights@email.com	03002223333	2024-05-12

Result Grid							
		Filter Rows:		Export:		Wrap Cell Content:	
	loan_id	book_id	member_id	staff_id	loan_date	return_date	status
▶	1	1	1	1	2025-01-05	2025-01-19	Returned
	2	2	2	2	2025-01-10	NULL	Issued
	3	3	3	1	2025-02-01	2025-02-15	Returned
	4	4	4	3	2025-02-10	NULL	Issued
	5	5	5	2	2025-03-01	NULL	Issued

9. Results and Conclusion

9.1 Summary of Tasks Completed

#	Task	Requirement	Status
1	Create Database & Tables	1 database, 5 related tables, sample data	Completed
2	Login Management	3 logins, password change, disable/enable	Completed
3	User Management	Users connected to database	Completed
4	Role Management	2 custom roles, users assigned	Completed
5	Permission Management	Grant, revoke, access demo	Completed
6	Backup & Recovery	Full backup, data deletion, restore	Completed

9.2 Conclusion

This lab assignment successfully demonstrated the core principles of database administration using MySQL Workbench. The Library Management System was designed with a clean relational schema consisting of five tables with properly enforced foreign key constraints. All six required tasks were completed:

- The database was created with normalized tables and meaningful sample data covering members, authors, books, staff, and loans.
- Three logins were created with password management and account locking demonstrated.
- Users were connected to the database and roles were defined to enforce access control.

- Permissions were granted to roles and users, with a revoke operation and access demonstrations showing both allowed and denied queries.
- A full backup was taken using mysqldump, data loss was simulated, and the database was successfully restored from the backup file.

The project demonstrates how role-based access control (RBAC) and privilege management are essential for maintaining database security, while proper backup procedures ensure data integrity and recovery in real-world systems.

9.3 Submission Checklist

- SQL Script File — library_management_system.sql
- Database Backup File — library_management_backup.sql
- Report with Screenshots — (this document)